CHARMM-GUI

Effective Simulation Input Generator and More

CHARMM is a versatile program for atomic-level simulation of many-particle systems, particularly macromolecules of biological interest. - M. Karplus

Some lectures, job postings, and FAQ are now available. See update log for update history and giving for donation. Contact info is given below.

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Bilayer Builder

User Profile

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PDB Info | STEP 1 | STEP 2 | STEP 3 | STEP 4 | STEP 5 | STEP 6

JOB ID: 3750189754

download.tgz

Original PDB File:	BTAAC1_CDL_CSTATE_2C3E.pdb (view structure)
Individual Chains:	btaac1_cdl_cstate_2c3e_proa.pdb btaac1_cdl_cstate_2c3e_heta.pdb
CHARMM Input:	step1_pdbreader.inp
CHARMM Output:	step1_pdbreader.out
CHARMM PDB:	step1_pdbreader.pdb (view structure)
CHARMM CRD:	step1_pdbreader.crd
CHARMM PSF:	step1_pdbreader.psf

Computed Energy:

Please beware of that the computed energy is CHARMM single-point energy and is displayed to make sure all the coordinates are defined.

```
ENER ENR:  Eval#     ENERgy      Delta-E         GRMS
ENER INTERN:          BONDs       ANGLes       UREY-b    DIHEdrals    IMPRopers
ENER CROSS:           CMAPs        PMF1D        PMF2D        PRIMO
ENER EXTERN:        VDWaals         ELEC       HBONds          ASP         USER
 ----------       ---------    ---------    ---------    ---------    ---------
ENER>        0  55042.97698      0.00000   4073.53398
ENER INTERN>      856.54129    789.62157    131.05931   3307.03701      2.81583
ENER CROSS>        -1.56780      0.00000      0.00000      0.00000
ENER EXTERN>    53502.27666  -3544.80689      0.00000      0.00000      0.00000
 ----------       ---------    ---------    ---------    ---------    ---------
```

Orientation Options: ?

○ Use PDB Orientation	This option is suggested for an oriented structure from http://opm.phar.umich.edu
○ Align the First Principal Axis Along Z	This option is suggested for small helical bundle or homo-oligomer.
○ Align a Vector (Two Atoms) Along Z	This option is suggested for an irregular, hetero-oligomer.
◉ Run PPM 2.0	This option run executable for given input structure at https://opm.phar.umich.edu/ppm_server2_cgopm It may take some minutes depending on protein size.

Please select the chains sending to PPM server.

☑ PROA

Positioning Options:

☐ Rotate Molecule respect to the X axis ____ Degree

☐ Rotate Molecule respect to the Y axis ____ Degree

☐ Translate Molecule along Z axis ____ Angstrom

☐ Flip Molecule along the Z axis

Area Calculation Options:

☑ Generate Pore Water and Measure Pore Size

- ◉ Using protein geometry   Water molecules outside the XY extent of the transmembrane domain would be removed.
- ○ Using a cylindrical radius of ______ Angstrom

Next Step:
Calculate Cross-Sectional Area

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